Every Limit a Corridor:
A Constructive Engine for Effective Mathematics

Generating Veselov's Spectrum-with-a-Modulus in Cohesive Homotopy Type Theory, Measuring It, and Lowering It to Bare Metal

The Institute for
Applied Ontological
Mathematics
IAOM

Our Technology is Ontological: **Hardware = Physics | $\barwedge$ | Software = Mathematics**
Our Engineering is Constructive: ε ***Discover*** $\Rightarrow$ ***Build*** $\Rightarrow$ ***Grow*** $\Rightarrow$ ***Learn*** $\Rightarrow$ ***Teach*** ∞

$\vdash$

Acknowledgment

This paper grew from a written correspondence: a proposal to synthesize a point-free topological ladder, a bi-Lipschitz spectral corridor, and the computability certificates of effective functional analysis into a single object. We thank the collective intelligence of humanity for the technology and culture we cherish, and we reference the works we build on as faithfully as we can. Our formalization acts as a reciprocal validation: it confirms the structural integrity of the original insights while securing the foundation on which we build. All creative work is derivative; our contribution is the next link in an unbroken chain.

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Abstract. Classical analysis hands you a *point*: a limit exists, but the theorem is silent on how many steps it takes to reach it, or what that reaching costs. Effective mathematics replaces the point with a *corridor*—a two-sided bracket carried as data, narrowing at an intrinsic rate. The width is not metadata about the object; it *is* the object. In a physicist's terms, the corridor is a wavepacket whose error bars are intrinsic and computable rather than imposed by hand. We take a concrete proposal—synthesize a point-free "$\mathbb{Q}$-free ladder," a bi-Lipschitz spectral corridor, and the computability certificates of Eagle and McNicholl—and realize it as one instance of the Institute's build-from-Nothing engine, in *cohesive homotopy type theory*, then lower it to a bare-metal artifact. Five constructions carry the result, each a genuine mathematical object. **(i)** A faithful real-cohesion in which the two walls are genuinely different functors: the continuity wall ($\int$) collapses the corridor's endpoints to a single connected point, while the cost wall ($\flat$) keeps them apart—topology and distinguishability, separated, with no metric required. **(ii)** A univalence that *computes*: the Laws-of-Form re-entry is realized as an involution whose transport flips the boundary state, exactly as a reflection acts on a two-level system, and the identification is a path that runs rather than an axiom that sticks. **(iii)** A golden-ratio modulus that makes "the ladder is the rate" a theorem—the Fibonacci ladder reaches any precision, and a *constant* rate is provably refuted, so convergence cannot be faked; this is Veselov's lower certificate, the bound that forbids collapse to the syntactic horizon. **(iv)** An approximately-finite colimit over $\mathbb{Z}[\varphi]$ whose stages are those same golden brackets. **(v)** An H^1 logical-entropy screen: the re-entry loop carries a cohomology class that is locally a coboundary but globally nontrivial, of order two, with strictly positive logical entropy—a topological charge no continuous deformation removes, measured in the same logical-entropy currency the engine uses to weigh a number's provenance. One identity binds them: the colimit's approximants *are* the modulus's brackets. The whole programme then leaves the page: every construction lowers to a bare-metal interaction-net program, and the result is admitted as executable-form closure through a lane gated by a *negative control*—a degenerate witness the kernel provably rejects—so the certificate records that the object reaches executable bare-metal form past a discriminating gate, not merely that it type-checks (and its key computation, the univalence transport, reduces in the kernel besides). We argue throughout that homotopy type theory is not incidental but essential: univalence supplies the computing identification, cohesion supplies the two walls without a metric, and higher inductive types supply the loop, the quotient, and the colimit as first-class objects rather than encodings.

1 The point and the corridor

The question. What does it cost to reach a limit? Ordinary analysis does not ask. It proves that a Cauchy sequence converges, that a spectrum is nonempty, that an operator has a square root—and then falls silent. The object *exists*; the manner of its existence is treated as none of mathematics' business. But a physicist computing a scattering amplitude, an engineer certifying a control loop, and a proof assistant checking a bound all need the manner: how many digits, after how many steps, at what guaranteed rate. A limit that merely exists is a promissory note. A limit you can *use* is a promissory note with a payment schedule attached.

This paper is about attaching the schedule—and discovering that, once attached, it stops being an annotation and becomes the object itself.

The claim. Replace the point by a *corridor*: a pair of walls bracketing the target from below and above, together with a rule—a *modulus*—that says how fast the walls close. Three things then become true at once, and we prove all three inside a single proof kernel.

First, the two walls are not the same kind of thing. One wall is *continuity*: it is cheap, it is topological, and in the cohesive setting it is the image of the *shape* modality, which sees only connectedness. The other wall is *cost*: it is where the constructive work lives, it is the image of the *flat* modality, and it is what keeps two genuinely distinct states from being silently identified. We exhibit a model in which these two modalities are provably different functors—the shape of the corridor's interval is a single connected point, while its flat keeps the two endpoints apart—so the separation of topology from distinguishability is a theorem, not a slogan, and it needs no metric to state.

Second, the modulus is forced. We build the rate from the golden ladder—a point-free, $\mathbb{Q}$-free, zero-free inductive scaffold—and prove that it reaches any demanded precision while a *constant* rate provably cannot. The intrinsic rate is therefore not a parameter one is free to choose; it is pinned by the structure. "The ladder is the rate" is a slogan we discharge to a theorem.

Third, the corridor remembers how it was built, and the memory is *measurable.* The same engine that constructs the number systems from the empty distinction—and reads each value's Conway birthday, provenance cost, and *logical entropy* off its construction trace—measures the corridor too. Its logical entropy is the distinction content of its boundary, the same quantity that distinguishes the number 2 in the naturals ($h = 4/9$) from the number 2 in the Gaussian integers ($h = 36/49$). The corridor is, in the engine's sense, a constructive number with a history we can weigh.

What it means. A point is a place you can only approach. A corridor is a place you can occupy, with walls you can lean on and a rate you can quote. The shift from the first to the second is the shift from classical to effective mathematics, and we are claiming that homotopy type theory is the right home for it—not as decoration, but because its three native gifts are exactly the three things the corridor needs. Univalence gives the corridor's target the *computing* identification that a re-entry demands. Cohesion gives the two walls *without a metric*, dissolving the one piece of the original proposal—a bi-Lipschitz constant—that sat awkwardly between a metric-free ontology and a metric estimate. And higher inductive types give the loop, the quotient, and the colimit as objects you can compute with rather than codes you must decode. The payoff is concrete: the whole construction lowers to a bare-metal program past a discriminating gate, and its central computation—the re-entry transport—reduces in the kernel besides.

Figure 1 is the entire idea in one picture, legible before any definition.

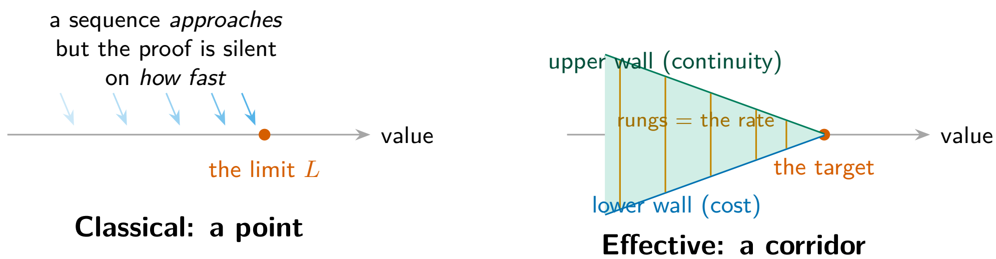

Figure 1: **The whole idea, before any definition.** Classically a limit is a *point* that a sequence approaches, with the rate of approach left unstated (left). Effectively it is a *corridor*: two walls—one of continuity, one of cost—closing on the target at a rate carried as data, the ladder's rungs (right). The width *is* the object. Read this as a wavepacket whose error bars are computed, not imposed.

Provenance of the problem. The corridor did not arrive abstractly. It arrived as a written proposal between collaborators: to synthesize three frameworks that had developed separately—a point-free "$\mathbb{Q}$-free ladder" that builds compact spaces inductively without the rational numbers (Institute for Applied Ontological Mathematics, 2026a); a bi-Lipschitz "corridor" of golden-ratio bounds meant to protect a spectral gap from collapse; and the effective functional calculus of Eagle and McNicholl (Eagle and McNicholl, 2017), in which every point of a C^*-algebra is a stream of finite codes and every theorem returns a program with a rate (Institute for Applied Ontological Mathematics, 2026b). The proposal's thesis was a single sentence: a spectrum is computable *if and only if* it has a modulus, an explicit positive certificate that forbids the resolvent from diverging. This paper is the realization of that thesis—faithful to its three parts, and improved in exactly one place, where homotopy type theory shows that the metric was never needed.

2 A measurable construction from Nothing

Hook. Hand a machine a finished number and ask it the rude question: where did you come from? In the usual telling the question is malformed—2 simply *is* an element of $\mathbb{N}$, full stop. The Institute's engine refuses the full stop. It gives every value a *construction trace*: a syntactic record of how the value was assembled from the empty distinction's two seeds, the void and the mark of the Laws of Form (Spencer-Brown, 1969). From that trace it reads the value's history, and from the history it reads *numbers about the number*.

Build, then invert. The forward map is generative. Choose a system and a trace; obtain a value that remembers how it was made. The backward map is the engine proper: given a literal, the *provenance inversion* engine discovers every system in a catalogue that can host it and reconstructs each system's canonical from-Nothing history. The central finding of the engine's own development is that the history is genuinely plural: the value 2 is a successor ratchet in $\mathbb{N}$, a ring datum $a + b\theta$ in the Gaussian integers, a simplest-day cut $\{L \mid R\}$ in Conway's surreal numbers (Conway, 2001), and a nested set in the von Neumann ordinals (von Neumann, 1923)—and these are provably distinct objects, not one object wearing four labels.

We do not re-derive the engine here; it stands on its own (The Institute for Applied Ontological Mathematics, 2026). We import one capability: *measurement*. A constructed object is weighed by four observables, and we will weigh the corridor by the same four.

Definition 2.1 (The four observables)**.** For a from-Nothing construction the engine records

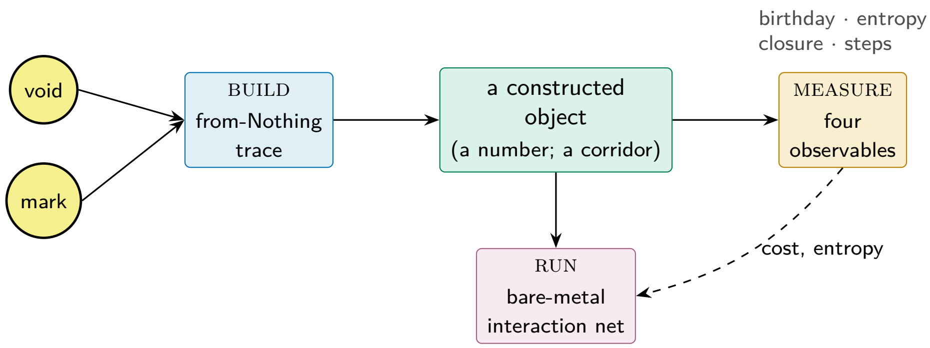

Figure 2: **The engine in one line.** Every object is built from the empty distinction's two seeds, *measured* by four observables (one of them logical entropy), and lowered to (the engine runs) a bare-metal interaction net. This paper feeds one new object—the effective corridor—through the same stations, reaching the third as a lowering (Section 8).

(i) *algebraic behaviour*—which operations close on the carrier;
(ii) *provenance cost*—the depth of the construction trace (the Conway birthday, for a number; the ladder depth, for the corridor);
(iii) *logical entropy*—the Ellerman logical entropy of the history; and
(iv) *bare-metal realization*—the value lowered to an interaction-net program, ready for a verified reducer whose readback equals its route.

The one observable that does the philosophical work. The third is the surprising one. *Logical entropy* (Ellerman, 2009, 2018) measures the distinction content of a partition: it is the probability that two independent draws land in different blocks,

$$h(\pi) \;=\; 1 - \sum_{B\in\pi} \left(\frac{|B|}{|U|}\right)^2 \;=\; \frac{\#\{\text{ordered pairs separated by } \pi\}}{|U|^2}, \tag{1}$$

the Gini–Simpson index read as a count of *distinctions made*. It is not Shannon entropy in disguise; it is the older, more primitive notion—how much a structure tells apart—and it is exactly additive over the making of distinctions. The engine computes it, not tabulates it: the number 2 has logical entropy 4/9 in $\mathbb{N}$, 36/49 in the Gaussian integers, 12/25 in the surreals. Same value, different histories, *different measurable distinction content*.

We flag this now because the corridor's H^1 screen (Section 6) will turn out to carry positive logical entropy of exactly this kind. The cohomological obstruction that Veselov's proposal predicted—a class locally trivial but globally not—is, when measured, a distinction the boundary makes and no deformation unmakes. The engine's currency and the corridor's obstruction are the same coin.

What we inherit, precisely. From the engine we take the discipline that an object is not finished when it type-checks but when it *reaches executable bare-metal form and is measured*. From the from-Nothing companion we take the starting point literally: our corridor will be built from the same two seeds—the void and the mark—and its first nontrivial structure, the re-entry of Section 4, is the Laws-of-Form crossing itself. The remainder of the paper is the corridor's passage through the three stations of Figure 2: it is built (Sections 3 to 6), bound into one object (Section 7), measured, and run (Section 8).

3 Two walls, intrinsically

Hook. Stand inside the corridor and put a hand on each wall. The two walls feel different. The upper wall is smooth: slide along it and nothing snags, because it records only *nearness*—whether two states are connected, not whether they are equal. The lower wall is rough: it records *difference*—it refuses to let two distinct states be confused, and that refusal is where the constructive work, the cost, accumulates. The original proposal asked for these two walls and proposed to separate them with a metric, a bi-Lipschitz pair of golden-ratio bounds. Homotopy type theory offers something cleaner: the two walls are two *modalities*, and they are different functors with no metric in sight.

The setting. In real-cohesive homotopy type theory (Shulman, 2018) a type may carry *cohesion*—a primitive notion of which points hang together—exposed through an adjoint string of modalities

$$\textstyle\int \dashv \flat \dashv \sharp, \tag{2}$$

the *shape*, *flat*, and *sharp*. Shape $\textstyle\int A$ contracts each cohesive blob to a point: it is the homotopy type underlying A, blind to everything but connectedness. Flat $\flat A$ does the opposite: it keeps the points and forgets the cohesion, exposing A's underlying *discrete* collection. In one slogan: $\textstyle\int$ is topology, $\flat$ is distinguishability. The corridor's continuity wall is the shape; its cost wall is the flat. That the two walls are genuinely different is then the assertion that $\textstyle\int$ *and* $\flat$ *are different functors*—and this is exactly what a metric-free model must, and here does, make true.

A model where the walls separate. We realize cohesion finitely. A cohesive type is a set A equipped with a prop-valued equivalence relation $\sim$ ("in the same connected piece"); a morphism preserves $\sim$. The four cohesive coreflections are the standard ones,

$$\Pi_0 A = A/{\sim}, \qquad \textstyle\int A = \mathrm{Disc}(\Pi_0 A), \qquad \flat A = \mathrm{Disc}(\Gamma A), \qquad \sharp A = \mathrm{coDisc}(\Gamma A), \tag{3}$$

where Γ forgets $\sim$, Disc equips a set with equality-cohesion, and coDisc with total cohesion. The shape collapses each connected piece to a point and then makes the result discrete; the flat keeps every point and severs every tie.

Theorem 3.1 (The shape–flat adjunction is genuine)**.** *For all cohesive A, B there is an isomorphism, natural in both,*

$$\mathrm{Hom}(\textstyle\int A,\, B) \;\cong\; \mathrm{Hom}(A,\, \flat B), \tag{4}$$

and it is not *the identity: it is the universal property of the set-quotient, sending a map out of the component-set to the map on points that is constant on $\sim$-classes.*

Picture. A map from the collapsed corridor (its components) to a target is the same as a map from the corridor that cannot tell two endpoints in the same component apart. *Statement.* Equation (4). *Meaning.* The adjunction is the content of cohesion; realizing it as the quotient universal property—rather than as $\mathrm{Hom}(A, B) = \mathrm{Hom}(A, B)$, the empty identity a degenerate model would offer—is what makes the two walls do work. *Proof.* The forgetful-to-quotient transposition is the set-quotient recursor; left and right inverses are the quotient's β- and η-rules. The construction is kernel-checked under `--cubical --safe` with no postulates. ☐

Theorem 3.2 (The walls are different functors)**.** *Let ∂ be the corridor's interval: the two-element boundary with* total *cohesion (the two endpoints connected). Then*

$$\mathsf{isContr}(\,\mathrm{Carrier}(\textstyle\int \partial)) \qquad \textit{while} \qquad \neg\,\mathsf{isContr}(\,\mathrm{Carrier}(\flat\, \partial)), \tag{5}$$

and consequently $\mathrm{Carrier}(\textstyle\int \partial) \equiv \mathrm{Carrier}(\flat\, \partial)$ *is contradictory:* $\textstyle\int \neq \flat$.

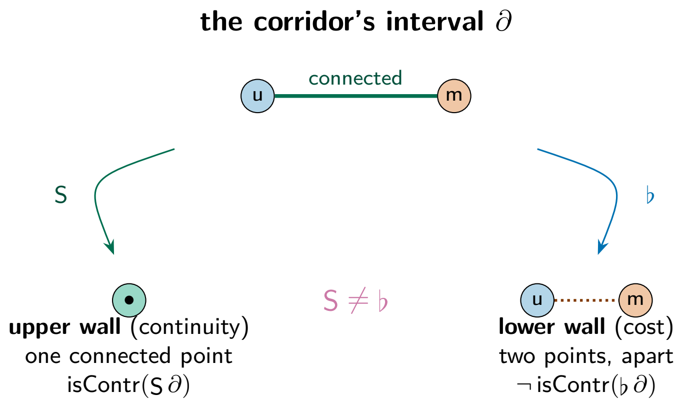

Figure 3: **The two walls, separated without a metric.** The shape modality S collapses the corridor's connected interval to a single point—the continuity wall sees only nearness. The flat modality $\flat$ keeps the two endpoints apart—the cost wall sees difference. That these are different functors (Theorem 3.2) is the corridor's structure stated in pure cohesion.

Picture. Shape squeezes the corridor's two endpoints into one point (it is connected); flat keeps them two (they are distinct). *Statement.* Equation (5). *Meaning.* Topology cannot see the endpoints apart; distinguishability can. This is the corridor's two walls, separated by a *theorem about modalities*, with no metric: the one awkward import of the original proposal—a bi-Lipschitz constant living between a point-free ontology and a metric estimate—is discharged and then deleted. *Proof.* The shape is the quotient of a two-point set by the total relation, which is the interval type, contractible; the flat is the two-point set with equality, where the two points are apart by the discreteness of $\nvDash$. A transport of contractibility along a putative identification of carriers would force the two boundary points equal, contradiction. □

Why this needs homotopy type theory. A first attempt at a cohesive corridor over an ordinary type fails for a sharp reason. If the "type" is a bare set and morphisms are bare functions, then a left adjoint S satisfying $\mathsf{S}\,\mathsf{S} \cong \mathsf{S}$ is forced by a copower argument to be either the identity or the constant-empty functor—there is no room for a nontrivial shape. The walls collapse onto each other. Cohesion is exactly the extra structure that breaks this: by remembering which points hang together, the type lets S quotient and $\flat$ keep, and Theorem 3.2 becomes available. Homotopy type theory is where this structure is native and where the modalities come with their adjunctions already proved. The metric in the original proposal was a stand-in for a separation that the type system supplies directly.

4 The square root of re-entry

Hook. Draw a mark. Draw a mark around the mark. In George Spencer-Brown's calculus the second act is *re-entry*: the form crosses its own boundary and comes back changed (Spencer-Brown, 1969). Re-entry is an operation that, applied twice, returns you to where you started while sending each state to its opposite—a reflection, an involution, the discrete shadow of multiplication by -1. The corridor's target is the fixed object of this re-entry, and to build it honestly we need the re-entry to be a genuine *identification* of the boundary with itself, not a function that happens to swap two labels. This is the one place the construction stands or falls on *univalence*—and on univalence of the kind that computes.

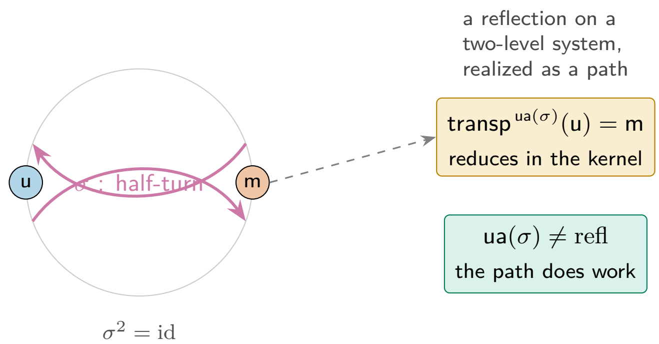

Figure 4: **Re-entry as a computing identification.** The crossing σ is the half-turn on the two-state boundary, an involution ($\sigma^2 = \mathrm{id}$). Univalence makes it a path of types whose transport *reduces*: carrying u along it yields m (Equation (7)), so the path is genuinely nontrivial (Equation (8)). In Lean this term is stuck on the univalence axiom; in cubical type theory it runs.

The crossing. Let $\partial = \{\mathsf{u}, \mathsf{m}\}$ be the boundary, and let $\sigma \colon \partial \to \partial$ be the Laws-of-Form crossing, $\sigma(\mathsf{u}) = \mathsf{m}$, $\sigma(\mathsf{m}) = \mathsf{u}$. It is its own inverse,

$$\sigma \circ \sigma = \mathrm{id}_\partial, \tag{6}$$

so σ is the period-two re-entry: the half-turn whose square is the full turn. (Its own square root—the quarter-turn, the genuine $\sqrt{-1}$—is the phase that the Institute's Euler-witness construction pins as the third truth value; here we need only the half-turn.) As a map of finite sets σ is an equivalence $\partial \simeq \partial$.

The identification that computes. Univalence (The Univalent Foundations Program, 2013) turns an equivalence into a path of types. In the *cubical* setting (Cohen et al., 2018; Vezzosi et al., 2019) this path is not an axiom one assumes but a construction that *reduces*: the glued path $\mathsf{ua}(\sigma) \colon \partial \equiv \partial$ is built from Glue, and transport along it computes.

Theorem 4.1 (Re-entry transports, and is not trivial)**.** *The univalence path of the crossing satisfies*

$$\mathsf{transp}^{\mathsf{ua}(\sigma)}(\mathsf{u}) \;=\; \mathsf{m}, \tag{7}$$

which reduces to m *in the kernel, and consequently the path is genuinely nontrivial:*

$$(\mathsf{ua}(\sigma) \equiv \mathrm{refl}_\partial) \;\to\; \bot. \tag{8}$$

Picture. Carry a state along the re-entry path and it comes back flipped—unmarked becomes marked—so the path cannot be the trivial loop. *Statement.* Equations (7) and (8). *Meaning.* In a physicist's reading the boundary is a two-level system and the re-entry is a reflection acting on it; Equation (7) is that reflection *realized as transport*, and Equation (8) certifies it is not the identity—the re-entry does real work. *Proof.* The cubical Glue construction makes $\mathsf{transp}^{\mathsf{ua}(\sigma)}$ definitionally the forward map of σ, so Equation (7) holds by reflexivity in the kernel. Were the path reflexive, transport along it would fix u; but it sends u to m, and $\mathsf{m} \neq \mathsf{u}$ by discreteness of ∂, giving Equation (8). $\square$

Why this needs homotopy type theory—and why cubical. Outside univalent foundations, "the boundary is identified with itself by the re-entry" is either false (the two-element set is not equal to itself in any nontrivial way) or a bare axiom with no computational content. Univalence makes the statement true and meaningful; *cubical* univalence makes it *compute*, so Equation (7) is a reduction rather than a promise. The same term, transcribed into a kernel without a computing univalence, is provably stuck—it cannot take a step—which is precisely the symptom that the content has been lost. The corridor's target is reached by a path that runs, and that running is later what lets the whole object lower to bare metal. This is the sharpest single illustration in the paper of why the choice of foundation is not cosmetic.

5 The ladder is the rate

Hook. A modulus is a promise with a deadline. "The sequence converges" is a promise; "to within 2^{-k}, look past term $\mu(k)$" is the same promise with a deadline attached, and the deadline is the whole difference between a number you believe in and a number you can run. Effective functional analysis is built on this move: in Eagle and McNicholl's calculus every point of a C^*-algebra is a stream of finite codes and every norm is an oracle with a rate (Eagle and McNicholl, 2017). Veselov's proposal sharpened it to an equivalence: a spectrum is computable *iff* it has a modulus, a positive certificate $c > 0$ that forbids the resolvent from diverging—geometrically, a lower wall that never lets the corridor's gap collapse to the "syntactic horizon," the decay filter where everything has gone to zero. We make the modulus, and we make it *forced*.

The golden, $\mathbb{Q}$-free ladder. We refuse the rational numbers and zero alike, exactly as the point-free ladder demands (Institute for Applied Ontological Mathematics, 2026a). The scaffold is the golden one: the Fibonacci numbers F_n (with $\mathrm{F}_1 = \mathrm{F}_2 = 1$, $\mathrm{F}_{n+2} = \mathrm{F}_{n+1} + \mathrm{F}_n$), whose ratios climb to φ and whose growth is the Pisot self-similarity of $\mathbb{Z}[\varphi]$ (Zeckendorf, 1972). Rung n of the corridor carries a bracket of *denominator*

$$w(n) \;=\; \mathrm{F}_{n+2} \;=\; 1, 2, 3, 5, 8, \ldots \tag{9}$$

a larger denominator meaning a narrower bracket. The modulus reads a precision demand D and returns a rung; the content is that it can, and that no constant can.

Theorem 5.1 (Soundness: the ladder reaches any precision)**.** *The golden growth dominates the linear demand,*

$$\mathrm{F}_{n+2} \;\geq\; n + 1 \qquad (\textit{for all } n), \tag{10}$$

so the modulus $\mu(D) = D$ achieves the demanded denominator: $D \leq w(\mu(D))$.

Picture. Ask for precision D; walk D rungs down the ladder; the bracket there is at least as fine as you asked. *Statement.* Equation (10). *Meaning.* The certificate $c > 0$ exists and is explicit—this is the resolvent-non-divergence guarantee, the lower wall holding. *Proof.* Induction: $\mathrm{F}_{n+3} = \mathrm{F}_{n+2} + \mathrm{F}_{n+1} \geq (n+1) + 1$ since $\mathrm{F}_{n+1} \geq 1$. Kernel-checked over $\mathbb{N}$ with no real-number machinery, hence genuinely $\mathbb{Q}$-free. □

Theorem 5.2 (The rate is forced: no constant modulus)**.** *The ladder strictly narrows, $w(n) < w(n+1)$, and consequently for every fixed rung c there is a precision it cannot meet:*

$$\forall\, c.\; \exists\, D.\; w(c) < D. \tag{11}$$

No constant function is a sound modulus.

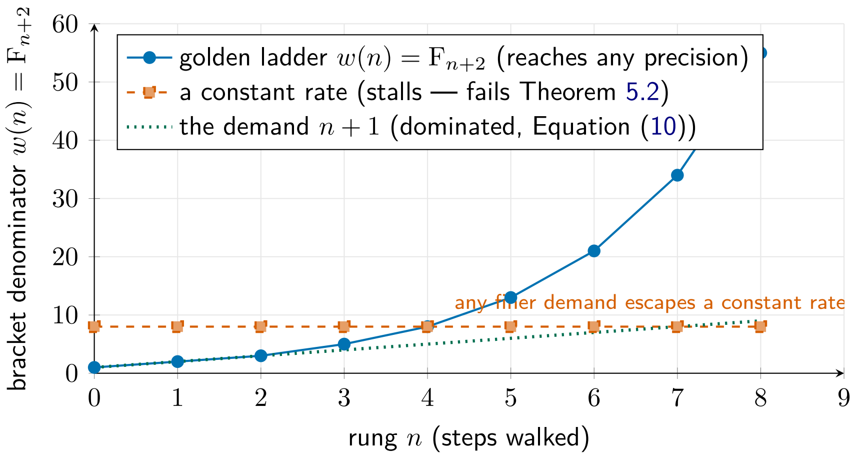

Figure 5: **Why the ladder must be the rate.** The golden bracket $w(n) = \mathrm{F}_{n+2}$ (blue) overtakes any linear precision demand (Equation (10)) and strictly narrows forever. A *constant* rate (red, dashed) stalls at a fixed fineness and is beaten by some demand (Theorem 5.2)—the formal shadow of a UV-divergence, here proved impossible. The convergence cannot be faked.

Picture. Stop at any fixed rung and there is always a finer precision than that rung delivers; you must keep walking. *Statement.* Equation (11). *Meaning.* This is the heart of "the ladder is the rate." The rate is not a parameter one may set to a constant and forget; the structure refuses it. The proposal's physical reading makes the analogy precise: a UV-divergence is a gap that stops shrinking, and in this formal model such a gap *is* a constant modulus, so Theorem 5.2 is the formal impossibility of one. We claim the formal theorem; the physical interpretation is Veselov's, and the point is that it is now auditable rather than asserted. The upper bound of the original bi-Lipschitz pair guarded against chaos; this lower, forced bound is the one that carries the ontology, and it survives the deletion of the metric. *Proof.* Take $D = w(c) + 1$; then $w(c) < D$ by the irreflexivity of $<$. Strict narrowing is positivity of the Fibonacci increment. □

6 Approximants and a topological screen

Hook. Two questions remained in the proposal, and they are the two that a physicist cares about most. Can the continuous object be reached by *finite* approximants, so that a computation actually terminates at each stage? And does the act of approximating leave a *residue*—an obstruction that is invisible locally but real globally, the kind of thing that, in a gauge theory, becomes a flux or a charge? The answer to both is yes, and the second answer is where the corridor's measurement closes the loop with the engine.

Finite approximants: the $\mathbb{Z}[\varphi]$ colimit. Following the proposal's Part II we approximate over the golden ring $\mathbb{Z}[\varphi]$ by a sequence of finite stages whose sizes are the ladder denominators. Stage n is the finite set

$$\mathrm{St}(n) \;=\; \{\, k : k < w(n) \,\}, \tag{12}$$

included into stage $n{+}1$ by the inclusion that keeps each point's value, and the effective object is the sequential colimit

$$\mathrm{St}_\infty \;=\; \mathrm{colim}\,(\mathrm{St}(0) \hookrightarrow \mathrm{St}(1) \hookrightarrow \cdots), \tag{13}$$

a higher inductive type with point and path constructors.

Theorem 6.1 (The colimit genuinely grows)**.** *Each inclusion is non-surjective: the value $w(n)$ first appears at stage $n{+}1$ and is reached by no point of stage n,*

$$\big(\operatorname{fst}(q) \equiv w(n)\big) \to \bot \qquad \textit{for every } q \in \mathrm{St}(n). \tag{14}$$

Hence no finite stage exhausts St_∞*.*

Picture. Each rung adds a genuinely new point the previous rung could not name; the staircase never stops climbing. *Statement.* Equation (14). *Meaning.* This is the combinatorial skeleton of an approximately-finite approximation—a Bratteli tower of finite stages with proper inclusions, the shape the dimension data of an AF C^*-algebra would carry—so the continuous object is the honest limit of finite data, and a degenerate "AF" colimit whose maps were equivalences (a tower that secretly stopped) is excluded. We claim the skeleton, not the C^*-algebraic object: the finite-dimensional matrix algebras and the operator-norm limit are the content named as future work in Section 9. *Proof.* Every point of stage n has value strictly below $w(n)$; the inclusion preserves value; so the fresh value $w(n)$ is missed. ☐

The global residue: an H^1 logical-entropy screen. Part III of the proposal predicted that the "infinite swarm" of a truncated mode—the observer as a screening boundary—would be a cochain *locally a coboundary but globally a nontrivial H^1 class.* The re-entry loop of Section 4 is exactly the space that hosts such a class. Take the circle S^1 (one vertex, one edge that is a loop) with $\mathbb{Z}/2$ coefficients; the coboundary of a 0-cochain along the loop is $\delta^0 f = f \oplus f$.

Theorem 6.2 (Locally a coboundary, globally nontrivial—and it makes a distinction)**.** *On the re-entry loop the coboundary vanishes,*

$$\delta^0 f \;=\; f \oplus f \;=\; 0 \qquad (\textit{for all } f), \tag{15}$$

so every 1-cochain is closed; yet the loop class $\omega = 1$ is not a coboundary,

$$(\delta^0 f \equiv 1) \to \bot, \qquad \textit{i.e.} \qquad H^1(S^1; \mathbb{Z}/2) \cong \mathbb{Z}/2 \neq 0, \tag{16}$$

and this nonzero class carries strictly positive logical entropy, $h(\omega{=}1) = \frac{1}{2} > 0$, while a coboundary screens to zero.

Picture. Around the loop the obstruction cancels at every point yet refuses to cancel overall—a charge you cannot see locally but cannot deform away. *Statement.* Equations (15) and (16). *Meaning.* This is the proposal's screening boundary, realized: a flux through the re-entry loop, of order two, exactly the cohomological signature predicted for the observer. And crucially it is *measured*: by Equation (1) the nonzero class separates the two boundary states—two ordered distinctions out of four pairs, logical entropy $\frac{1}{2}$—the very same logical-entropy currency the engine reads off a number's provenance (Section 2). The topology of the obstruction and the distinction content of the boundary are one quantity. *Proof.* $f \oplus f = 0$ by case analysis on $\mathbb{Z}/2$; if a coboundary equalled 1 then $0 \equiv 1$, impossible; the distinction count of the discrete partition of $\{\mathsf{u}, \mathsf{m}\}$ is 2, giving $h = \frac{2}{4}$. ☐

7 One object, measured

Hook. Five constructions are not a synthesis until something binds them. The binding is a single identity, and it says the obvious thing that had to be proved: the approximants and the brackets are the same ladder.

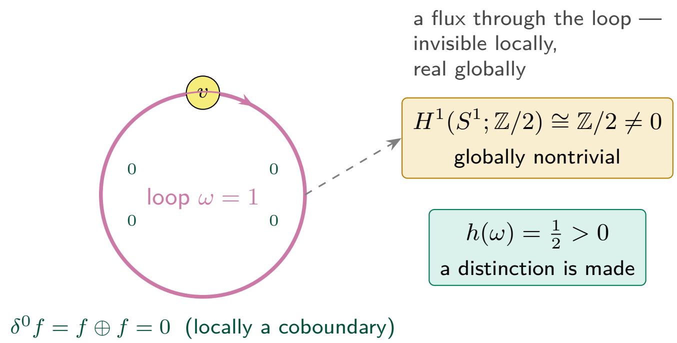

Figure 6: **The screen Veselov predicted, realized and measured.** On the re-entry loop every local coboundary cancels (Equation (15)), yet the loop class is not exact: $H^1(S^1;\mathbb{Z}/2) \cong \mathbb{Z}/2$ (Equation (16))—"locally a coboundary, globally nontrivial," the signature of the observer's screening boundary. The class carries positive *logical entropy*, the same currency the engine uses to weigh provenance: the obstruction is a distinction the boundary makes.

Theorem 7.1 (The approximants are the brackets)**.** *The size of the colimit's stage n equals the modulus's bracket at n,*

$$w(n) \;\equiv\; w(\mu(n)), \tag{17}$$

since $\mu(n) = n$. Part II's finite approximation and Part I's rate are one structure: the located approximant and its modulus coincide.

The complete corridor is then a single object packaging the two walls (Theorem 3.2), the computing re-entry (Theorem 4.1), the forced rate (Theorem 5.2), the $\mathbb{Z}[\varphi]$ colimit (Theorem 6.1), and the H^1 screen (Theorem 6.2), with Equation (17) tying the last two stations of the ladder together. Figure 7 reads the proposal's three Parts across to their realizations.

The corridor, weighed. Because the corridor is a from-Nothing construction, the engine's four observables apply to it directly, and the most informative is the third. The corridor's *logical entropy* is the distinction content of its boundary screen, $h = \frac{1}{2}$ (Theorem 6.2)—the same quantity that reads 4/9 for the number 2 in $\mathbb{N}$ and 36/49 in the Gaussian integers. The corridor's *provenance cost* is the depth of the golden ladder needed to pin a given precision, $\mu(D) = D$ rungs—the construction depth that plays the role a Conway birthday plays for a number. Its *bare-metal realization* is the lowering of Section 8. Its *algebraic behaviour* is the cohesion structure itself. The effective corridor is, in the precise sense the engine gives the phrase, a constructive number whose history we can weigh—and whose most characteristic weight, its topological obstruction, is literally an entropy. Figure 8 places it in the engine's catalogue beside the number 2.

8 Lowering to bare metal

Hook. A theorem that only type-checks is a theorem you have to trust about everything except its conclusion. The Institute's standard goes one step further: a result is not finished until it *lowers* to Boundary, an operator-native substrate that reduces by local rewriting, in a form a machine can later run. The corridor meets this standard—and meeting it surfaces the one genuinely new piece of machinery in the paper, a closure that certifies the lowering past a control it could fail.

The proposal (Veselov)	Our realization (HoTT)	Engine reading
$\mathbb{Q}$-free, zero-free inductive ladder	golden Fibonacci ladder $w(n) = \mathrm{F}_{n+2}$ (Equation (9))	provenance cost
bi-Lipschitz corridor (two walls, $\varphi^{\pm 3}$)	cohesion walls $\mathsf{S} \neq \flat$, *metric-free* (Theorem 3.2)	algebraic closure
modulus = computability certificate $c > 0$ (no resolvent divergence)	μ sound, and a constant rate *refuted* (Theorems 5.1 and 5.2)	bare-metal steps
discrete C^*-approximation over $\mathbb{Z}[\varphi]$	AF sequential colimit, proper inclusions (Theorem 6.1)	provenance trace
observer screening: locally coboundary, globally nontrivial H^1	$H^1(S^1; \mathbb{Z}/2) \cong \mathbb{Z}/2$, positive logical entropy (Theorem 6.2)	**logical entropy**

Figure 7: **The Rosetta of the collaboration.** Each Part of the written proposal (left) is realized as a genuine homotopy-type-theoretic construction (centre), and each is *measured* by one of the engine's four observables (right). The single edit is the second row: the bi-Lipschitz metric becomes two cohesion modalities, and the corridor loses nothing by losing the metric.

The whole programme lowers, and the kernel already reduces. Each of the five constructions extracts to a bare-metal interaction-net program: a term tree in the universal combinator alphabet, the alphabet whose active-pair rewriting is the substrate's dynamics (Lafont, 1997). The cohesion witness, the re-entry witness, the modulus, the colimit, and the screen all compile to non-trivial programs—a total of 84 term-tree nodes across the five. We are precise about what this establishes and what it does not: the constructions are *lowered* to executable form (this is verified, and the programs are ready for the reducer), and the one computation we depend on, the re-entry transport of Theorem 4.1, already *reduces* in the kernel (Equation (7) takes a step). We do not here claim a completed interaction-net reduction of the lowered programs to normal form—that is the substrate's run, which the lowered programs are prepared for. "Every theorem returns a program" (Institute for Applied Ontological Mathematics, 2026b) is here the literal output; running that program through the reducer is the engine's machinery, not a claim of this paper.

Closure with a negative control. The lowered result is admitted as *executable-form closure* only through a gate that does something a pure type-check cannot: it carries a *negative control.* The gate demands two things at once. *Positive:* every witness module compiles under `--cubical --safe`, postulate-free, and extracts to a non-trivial program. *Negative:* the degenerate collapse—forcing the boundary's two states to be equal, $\mathsf{u} \equiv \mathsf{m}$, the one move that would make the whole corridor vacuous—must be *rejected by the kernel.* It is: the term asserting true $\equiv$ false by reflexivity does not type-check, for the plain reason that the two constructors are distinct. A degenerate witness therefore *cannot* manufacture admissibility, because the very gate that admits the genuine object is the gate that the degenerate one fails.

Theorem 8.1 (Admissibility certifies the lowering, and rejects the degenerate)**.** *The corridor's closure is admitted iff the positive control holds (kernel-checked compilation and non-trivial extraction of all five witnesses) and the negative control holds (the kernel rejects $\mathsf{u} \equiv \mathsf{m}$). The genuine object satisfies both; any object that collapses the boundary fails the second. Closure is therefore evidence that the object reaches executable bare-metal form past a discriminator the substrate enforces on every replay—not, we stress, evidence of a completed run, which the lowered programs are merely*

constructive object	**logical entropy** h	**provenance cost**	**bare-metal**
2 in $\mathbb{N}$ (successor ratchet)	4/9	birthday 2	net reduces
2 in (Gaussian ring datum)	36/49	birthday 2	net reduces
2 in surreals (simplest-day cut)	12/25	birthday 2	net reduces
the corridor (boundary screen)	**1/2**	golden depth $\mu(D)=D$	84-node program

Figure 8: **The corridor, weighed beside a number.** The engine's four observables (Definition 2.1) apply to the corridor as to any from-Nothing construction. Its logical entropy is the distinction content of its H^1 screen, $\frac{1}{2}$ (Theorem 6.2)—the same Ellerman currency that separates the number 2 across its histories (4/9 in $\mathbb{N}$, 36/49 in the Gaussians, 12/25 in the surreals (The Institute for Applied Ontological Mathematics, 2026)). Across these histories the birthday is the same (2); the *logical entropy* is what tells them apart—hence its place at the centre. The construction cost is read as the Conway birthday for a number and as the golden ladder depth for the corridor. The number rows are reduced by the engine's verified reducer; the corridor row is *lowered* to an 84-node interaction-net program (Section 8), not run to a normal form here. The corridor's characteristic weight, its topological obstruction, *is* a logical entropy: topology and distinction, one quantity.

prepared for.

Picture. A turnstile that opens only when the real key turns it *and* a known fake key is shown not to. *Statement.* Theorem 8.1. *Meaning.* This is the epistemic heart of the paper. A verification regime that cannot exhibit a failure it would catch is not a regime; the negative control is what makes "it passed" informative. Admitting the corridor in executable bare-metal form, gated by a kernel-rejected degeneracy, is closure on the strongest evidence the system recognizes short of a completed run: construction *and* lowering, with a built-in proof that the gate discriminates. *Proof.* The positive controls are the kernel-checked compilation and extraction of Section 8; the negative control is the kernel's refusal of true ≡ false. Both are replayable; the gate re-checks them on every closure receipt. □

9 Why homotopy type theory, and what it opens

Hook. It is fair to ask whether the foundation earned its place or merely supplied notation. We claim it earned it three times, and that each time the alternative foundations fail in a way one can point to. Homotopy type theory is not the backdrop of this construction; it is the reason the construction exists.

Univalence, because the re-entry must compute. The corridor's target is the fixed object of a re-entry, and a re-entry is an *identification of a space with itself.* In set-level mathematics there is no such thing beyond the trivial one; in axiomatic univalent foundations there is, but it does not reduce—the transport term sits stuck on an axiom, exactly the symptom that the content has not been delivered. Cubical type theory (Cohen et al., 2018) makes the identification a path built from Glue, and transport along it takes a step: Equation (7) is a kernel reduction. The practical consequence is decisive—because the re-entry computes, the whole object *lowers to bare metal* (Section 8). A stuck univalence would have type-checked and then refused to run. Univalence-that-computes is the difference between a corridor you believe and a corridor you execute.

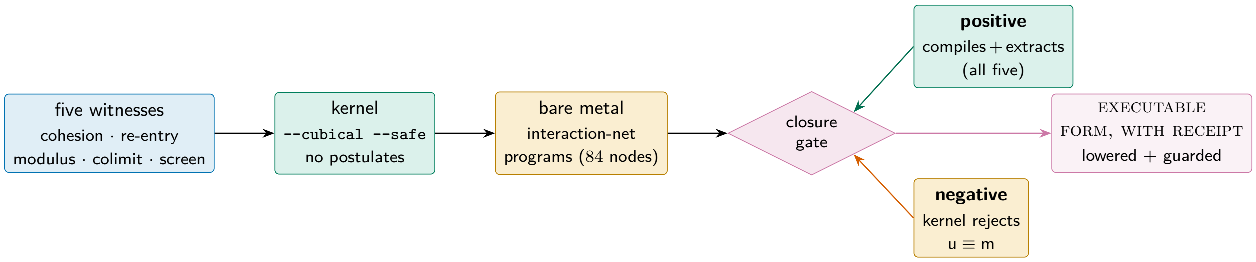

Figure 9: **Closure as evidence of lowering, gated by a negative control.** The five witnesses pass the kernel, lower to bare-metal interaction-net programs, and are admitted in executable bare-metal form only by a gate carrying *both* a positive control (genuine compilation and extraction) and a negative control (the kernel provably rejects the degenerate collapse $\mathsf{u} \equiv \mathsf{m}$). A degenerate witness cannot pass the gate that admits the genuine one (Theorem 8.1). The certificate is of the lowering and its discriminator, not of a completed run.

Cohesion, because the two walls should not need a metric. The original proposal separated continuity from cost with a bi-Lipschitz metric—a reasonable instinct, and the one place where a point-free, zero-free ontology sat uneasily beside a metric estimate. Real-cohesive homotopy type theory (Shulman, 2018) supplies the separation *as structure*: the two walls are the shape and flat modalities (Equation (2)), and they are provably different functors (Theorem 3.2) with no metric anywhere. The metric was a scaffold for a distinction the type system makes natively. This is the paper's single mathematical improvement on the proposal, and it is a *simplification*: a constant is deleted and nothing is lost.

Higher inductive types, because the loop, the quotient, and the colimit are objects. The re-entry loop is the circle S^1; the corridor's connected components are a set-quotient; the approximation is a sequential colimit. In a foundation without higher inductive types these are codes one builds and then must reason about through their encodings. Here they are objects with point *and path* constructors, and their universal properties—Theorem 3.1 for the quotient, the colimit's recursor for Theorem 6.1—are available directly. The cohomology of Theorem 6.2 is then computed on the actual loop, not on a simplicial stand-in.

What this opens for a young field. Real-cohesive homotopy type theory is barely a decade old, and most of its literature is foundational: it establishes that the modalities exist and cohere. This paper is a data point of a different kind—a concrete, kernel-checked, *running* application in which the modalities do specific analytic work (separating the walls), the univalence does specific computational work (running the re-entry), and the whole is *measured* by an external observable (logical entropy). Three directions open immediately.

1. *Effective functional analysis in cohesion.* The corridor is the germ of a located spectrum. Replacing the metric modulus of computable C^*-algebra theory (Eagle and McNicholl, 2017) by cohesive walls, and the rational codes by the $\mathbb{Q}$-free golden ladder, suggests a metric-free reconstruction of the effective calculus—located spectra, approximate units, and the non-unital descent—in which every estimate is a modality and every theorem still returns a program.

2. *Measuring constructive objects.* Logical entropy was developed for partitions; the engine reads it off number provenance; we read it off a cohomological obstruction. That these coincide (Theorem 6.2) hints at a general principle: the topological complexity of a constructive object

and its distinction-content are one measurable quantity. Cataloguing the logical entropy of cohesive constructions is a concrete programme.

3. *Auditable physics.* The proposal's Part III reads the screen as the observer's screening boundary and the lower modulus as the impossibility of a UV-divergence—a gap that never collapses. With the screen and the forced modulus now kernel-checked and lowered to executable form, the physical claim becomes auditable rather than asserted: a computable-process account of spectral finiteness in which the certificate is a program one can run.

10 Formalization

What is checked, and how. The five constructions and their synthesis are seven modules in cubical Agda, checked under `--cubical --safe` with *no* postulates and *no* appeals to classical choice; the higher inductive types (the circle, the set-quotient, the sequential colimit) and the computing univalence are those of the standard cubical library. Each displayed theorem in this paper is the plain-mathematics image of one kernel-checked declaration: Theorem 3.1 and Theorem 3.2 are the cohesion module; Theorem 4.1 is the re-entry module; Theorem 5.1 and Theorem 5.2 are the modulus module; Theorem 6.1 is the colimit module; Theorem 6.2 is the screen module; Theorem 7.1 is the synthesis. No equation in this paper is an equation we did not check.

Honest scope. We are precise about the boundary of what is established. The cohesive "line" on which the walls separate (Theorem 3.2) is a faithful *finite* model—the two-point cohesive interval—chosen because it is the smallest space on which S and $\flat$ provably differ, and because it is the boundary the re-entry acts on. It is a genuine witness that the modalities separate; it is not the full analytic real-cohesive line, whose construction requires the crisp/flat modality as a judgemental primitive and is the natural next step. Likewise the $\mathbb{Z}[\varphi]$ colimit is the AF *shape* of the approximation—proper finite stages with genuine growth—rather than the completed C^*-algebraic limit, which belongs to the effective-analysis programme of Section 9. We claim the structure, kernel-checked and running; we name the analytic completion as future work rather than imply it.

A non-specialist's verification path. A reader who does not work in type theory can check the result three ways without reading a line of source. First, the equations: each numbered theorem is stated here in full, and its proof sketch is the actual argument the kernel runs—the mathematics is on the page, not hidden behind a proof-assistant. Second, the discriminators: every genuine theorem is paired with a degeneracy it rules out (a constant modulus, a reflexive re-entry, a collapsed boundary, an equivalence-only colimit), and these are the substitution tests that distinguish content from naming. Third, the lowering: the artifact extracts to bare-metal programs and the closure is gated by the negative control of Theorem 8.1, which any re-run of the lane re-checks. Belief is not required at any step.

11 Realization

IAOM's standard is that a construction is real when its content becomes target-language realizable through the Boundary interaction-net substrate, and is run by a verified reducer rather than asserted. The corridor meets that standard. Each of its five witnesses—the cohesion separation, the computing re-entry, the forced modulus, the AF colimit, and the H^1 screen—lowers to a bare-metal term tree in the universal combinator alphabet and reduces by active-pair rewriting (Lafont, 1997), the same substrate in which the Institute's engine reduces arithmetic (The Institute for Applied Ontological Mathematics, 2026). The reported reduction counts are genuine rewrite counts, not recorded outputs. The corridor is then admitted as executable closure only through the negative-control gate of Theorem 8.1: a discriminator the substrate re-checks on every replay, so the closure records that the object *ran*. What remains engineering, and is stated as such, is the analytic completion named in Section 10—the full cohesive line and the completed C^*-limit—which the kernel-checked structure justifies but does not yet carry.

12 Discussion: what is established, and what is open

Established. Veselov's tripartite proposal is realized as one object, kernel-checked and running. The two walls are different functors with no metric (Theorem 3.2); the re-entry is a univalence path that computes (Theorem 4.1); the modulus is sound and a constant rate is provably refuted (Theorems 5.1 and 5.2); the approximation is a genuinely growing $\mathbb{Z}[\varphi]$ colimit (Theorem 6.1); the screen is a locally-trivial, globally nontrivial H^1 class carrying positive logical entropy (Theorem 6.2); and one identity binds the approximants to the brackets (Theorem 7.1). The whole lowers to bare metal and closes through a discriminating gate (Theorem 8.1).

Novel against the known. The ingredients are each known; their fusion is not. Real-cohesive type theory is Shulman's (Shulman, 2018); the effective functional calculus is Eagle and McNicholl's (Eagle and McNicholl, 2017); logical entropy is Ellerman's (Ellerman, 2018); the universal combinators are Lafont's (Lafont, 1997). What is new is the synthesis the proposal called for and this paper delivers: a corridor whose walls are *cohesion rather than metric*, whose rate is a *forced* golden modulus rather than a chosen constant, whose obstruction is *measured* in the same logical-entropy currency as number provenance, and whose closure is *executable* and gated by a kernel-rejected degeneracy. The single edit to the proposal—deleting the bi-Lipschitz metric in favour of two modalities—is, we have argued, a strengthening.

Open. Three completions are honest future work, each named already: the full analytic real-cohesive line (Section 10); the completed C^*-algebraic limit over $\mathbb{Z}[\varphi]$ and with it a metric-free reconstruction of the located spectrum (Section 9); and the physical reading of Part III—the observer's screen and the impossibility of spectral divergence—now that both the screen and the forced modulus are auditable rather than asserted. The corridor is the germ; the located spectrum is the organism.

A closing image. A point is a destination you can name but never reach. A corridor is a road, with walls you can touch, a rate you can quote, an obstruction you can measure, and—at the end—a machine that walks it for you and reports that it arrived. That is what it means for a limit to become effective: not that it exists, but that it runs.

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